

THE GAUSSIAN ROW–COLUMN EQUIVALENCE IN ARXIV:1203.3713 FOLLOWS BY SYMMETRIC BLOCK DILATION

AGENT_LANE_03

STATUS

Status: literature_implied_answer (full Gaussian question). The implication below completely proves the Gaussian equivalence suggested by Riemer–Schütt. It is not claimed as a new result: it is a direct reformulation of a later theorem of Latała–van Handel–Youssef.

SOURCE QUESTION

Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ be deterministic, let (g_{ij}) be independent standard Gaussian variables, and put $G = (a_{ij}g_{ij})$. Define

$$R(G) = \max_i \left(\sum_j G_{ij}^2 \right)^{1/2}, \quad C(G) = \max_j \left(\sum_i G_{ij}^2 \right)^{1/2}.$$

On page 3 of arXiv:1203.3713, immediately after display (2), Riemer and Schütt write that they know no matrix for which $\mathbf{E}\|G\|_{\text{op}}$ is not of the same order as that display, namely

$$(1) \quad \mathbf{E}R(G) + \mathbf{E}C(G).$$

Thus the suggested dimension-free equivalence is

$$\mathbf{E}\|G\|_{\text{op}} \asymp \mathbf{E}R(G) + \mathbf{E}C(G),$$

with universal constants, uniformly over n and A .

SUPPORTING THEOREM

Latała–van Handel–Youssef prove the following in [2, Theorem 3.1, p. 16]; it is the probabilistic form of their main Theorem 1.1. If X is a symmetric matrix whose upper-triangular entries are independent centered Gaussian variables, then, for every $2 \leq p \leq \infty$,

$$\mathbf{E}\|X\|_{S_p} \asymp \mathbf{E} \left[\left(\sum_i \left(\sum_j X_{ij}^2 \right)^{p/2} \right)^{1/p} \right].$$

At $p = \infty$ this reads

$$(2) \quad \mathbf{E}\|X\|_{\text{op}} \asymp \mathbf{E} \max_i \left(\sum_j X_{ij}^2 \right)^{1/2},$$

with universal constants.

IDENTIFICATION AND COMPLETE DEDUCTION

Form the symmetric $2n \times 2n$ block dilation

$$X = \begin{pmatrix} 0 & G \\ G^\top & 0 \end{pmatrix}.$$

The entries in the upper-right block are precisely the independent Gaussian variables $a_{ij}g_{ij}$; all other upper-triangular entries are deterministic zeros. Hence X satisfies the hypotheses of (2). Moreover,

$$X^2 = \begin{pmatrix} GG^\top & 0 \\ 0 & G^\top G \end{pmatrix}, \quad \|X\|_{\text{op}} = \|G\|_{\text{op}},$$

and inspection of its two row blocks gives

$$\max_{k \leq 2n} \left(\sum_{\ell \leq 2n} X_{k\ell}^2 \right)^{1/2} = \max\{R(G), C(G)\}.$$

Applying (2) therefore yields

$$(3) \quad \mathbf{E}\|G\|_{\text{op}} \asymp \mathbf{E} \max\{R(G), C(G)\}.$$

Finally, pointwise,

$$\max\{R, C\} \leq R + C \leq 2 \max\{R, C\}.$$

Taking expectations in this inequality and combining with (3) proves

$$\boxed{\mathbf{E}\|G\|_{\text{op}} \asymp \mathbf{E}R(G) + \mathbf{E}C(G)}.$$

This is exactly the equivalence suggested after display (2) of arXiv:1203.3713.

PROVENANCE, SCOPE, AND SEARCH EVIDENCE

The supporting authors explicitly state that their $p = \infty$ theorem settles Latała’s symmetric Gaussian conjecture. In the inspected source, they do not explicitly identify the Riemer–Schütt row–column formulation; the rectangular statement is obtained here by the standard block-dilation identification. The correct status is therefore `literature_implied_answer`, not a new proof of the run.

The result covers every deterministic coefficient matrix with independent standard Gaussian multipliers. It does not settle the source paper’s separate estimate for arbitrary independent mean-zero entries. The bounded status search covered the run indexes for arXiv:1203.3713 and the phrases “Gaussian matrix”, “row norm”, “column norm”, and “Latała”, then checked the exact arXiv sources 1203.3713 and 1711.00807. A prior run packet for Latała’s symmetric conjecture led to the decisive Theorem 3.1.

REFERENCES

- [1] S. Riemer and C. Schütt, *On the Expectation of the Norm of Random Matrices with Non-Identically Distributed Entries*, arXiv:1203.3713 (2012), <https://arxiv.org/abs/1203.3713>.
- [2] R. Latała, R. van Handel, and P. Youssef, *The dimension-free structure of nonhomogeneous random matrices*, Invent. Math. **214** (2018), 1031–1080; arXiv:1711.00807, <https://arxiv.org/abs/1711.00807>.